Slow flavor conversion in core-collapse supernovae

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N3AS News Session, Tuesday, May 12, 2026 — Online Seminar

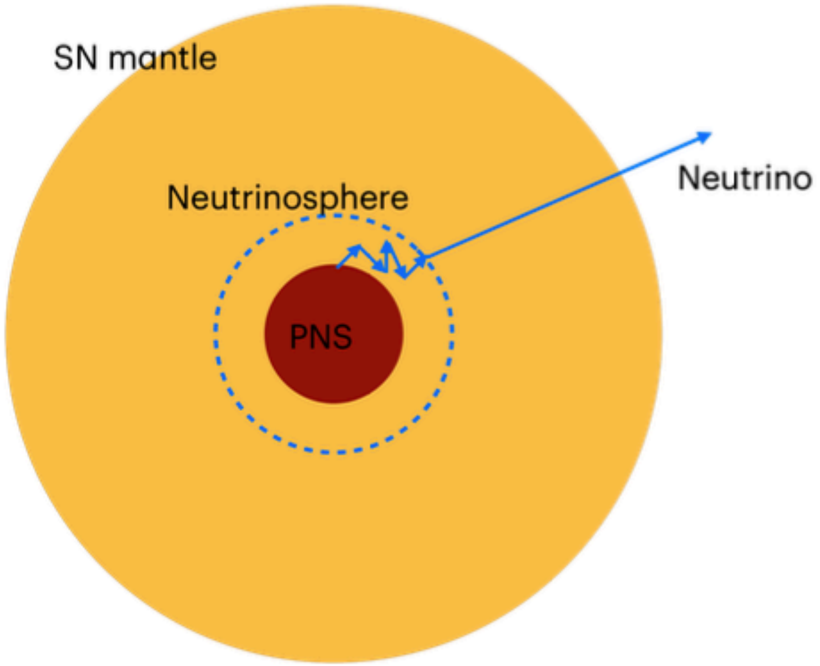
N3AS PHYSICS FRONTIER CENTER

Berkeley UNIVERSITY OF CALIFORNIA

UC San Diego

Image credits: T. Janka (left), S. Rosswog (right)

Neutrinos from core-collapse SNe



- **Energy scale:** 99% of the released energy ($\sim 10^{53}$ erg) is emitted in the form of ν and $\bar{\nu}$ of all flavors, with $E \sim 15$ MeV
- **Time scale:** ~ 10 s
- **Expected:** 1-3 SN/century in our galaxy ($d \approx 10$ kpc)

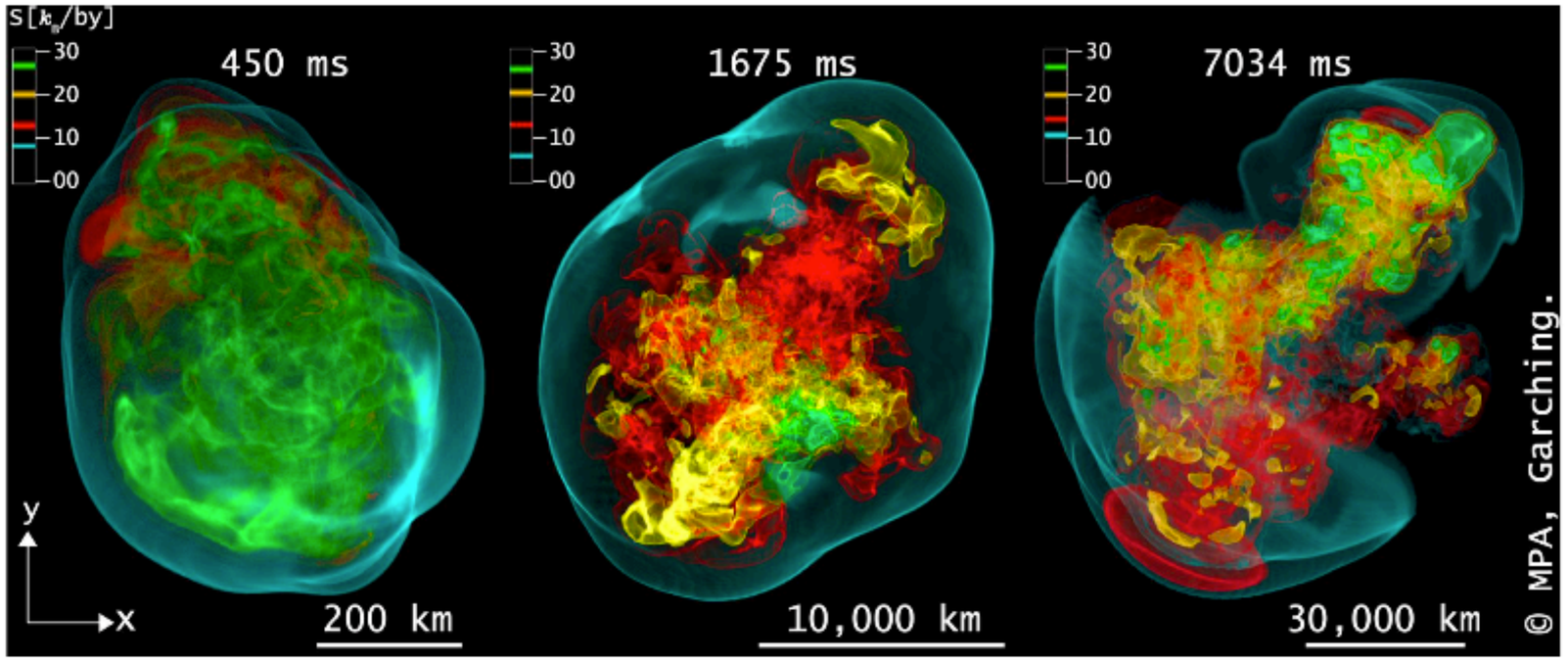
(To compare, the Sun will radiate $\sim 10^{51}$ erg over 10 billion years).

CCSNe are the most efficient neutrino factories in the Universe (10^{58} ν's)

SN simulations

Most of our knowledge about SN explosions is based on complex 3D simulations:

1. Account for the effect of convection and turbulence
2. Account for neutrino transport in 3D. No detailed treatment of flavor conversion (yet)

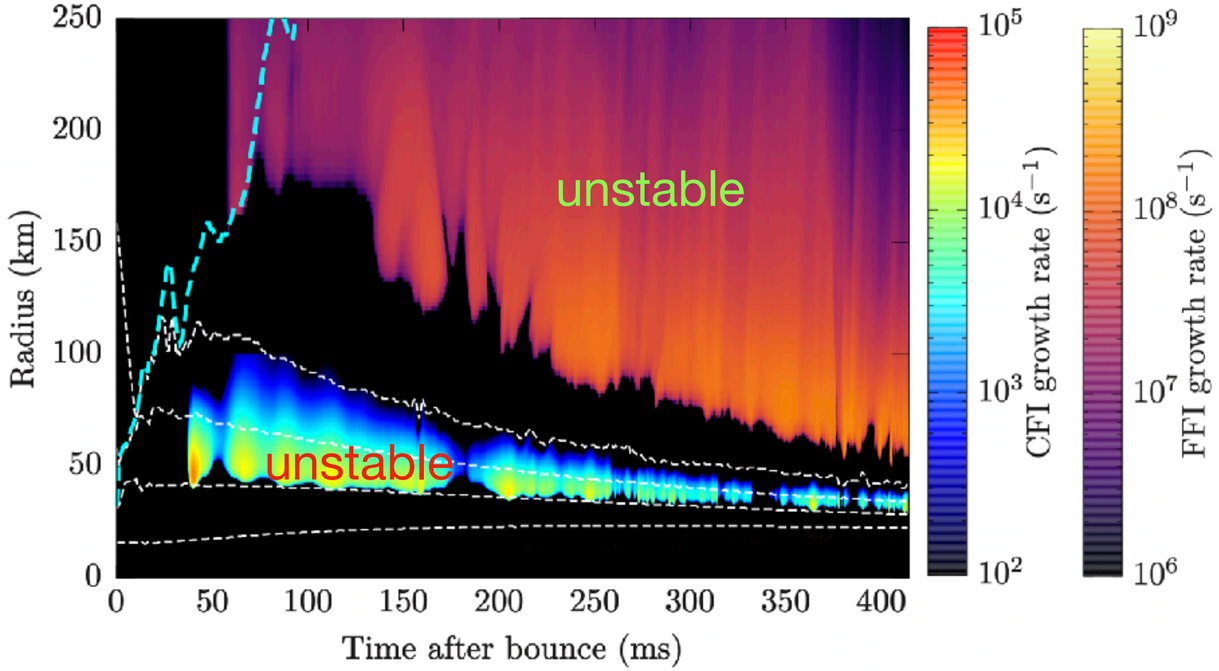


R. Bollig et al, 2010.10506

Flavor oscillations are expected

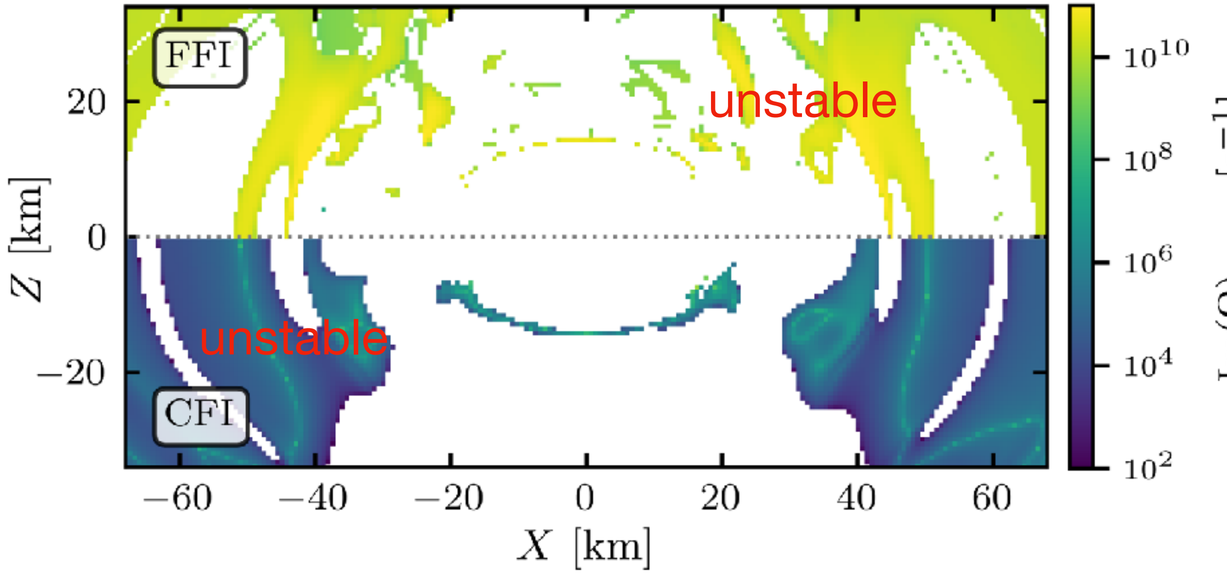
- Look for *flavor instabilities* in SN and NS merger simulations
- Linear stability analysis → unstable solution → flavor oscillations expected

SN simulation



R. Akaho et al, 2311.11272

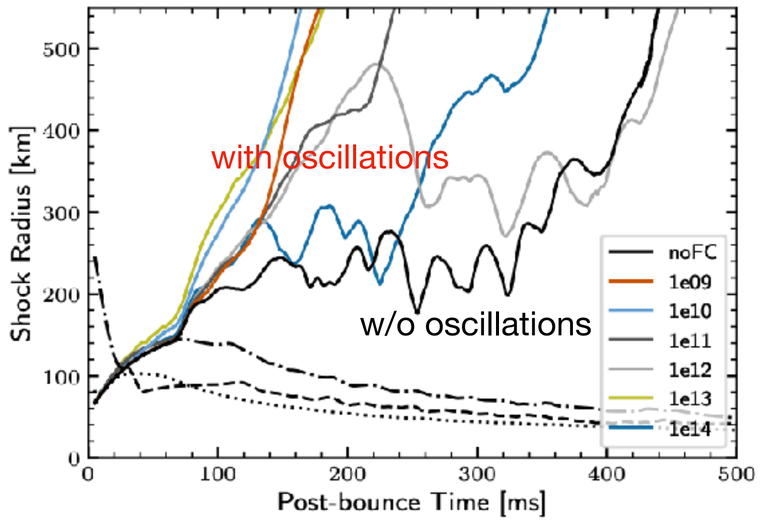
NS merger simulation



J. Froustey, 2311.11968

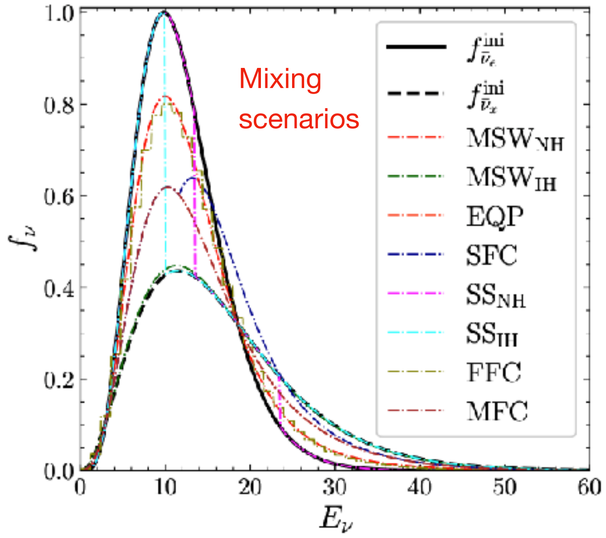
Potential impact of neutrino oscillations

Supernova dynamics



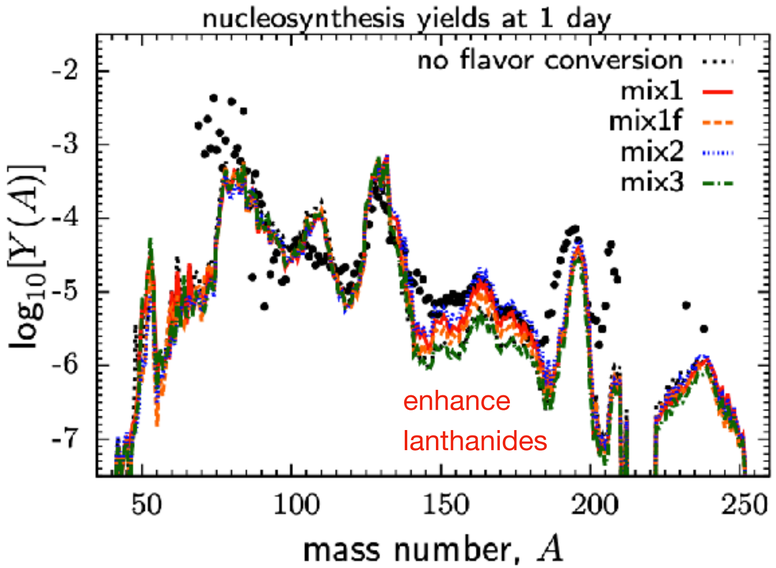
J. Ehring et al, 2305.11207

ν energy spectra



S. Abbar et al, 2401.10851

r-process nucleosynthesis



O. Just et al, 2203.16559

People are estimating the influence of oscillations on:

- **Proton-rich nucleosynthesis** in SN: νp-process (A. Friedland, IPG et al, 2508.02055)
- **Gravitational-wave emission** in SN (J. Ehring et al, 2412.02750)

Annual Review: L. Johns et al, 2503.05959

Current challenges

Supernova dynamics

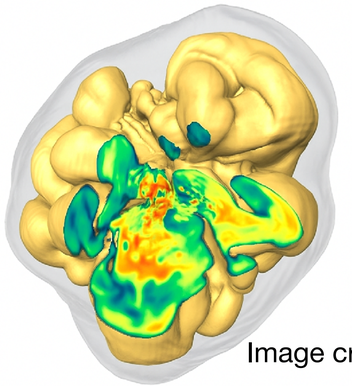
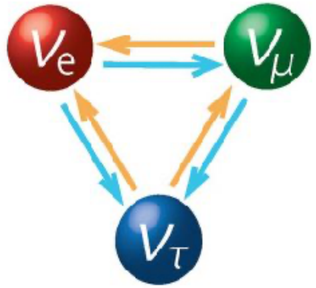


Image credits: T. Janka

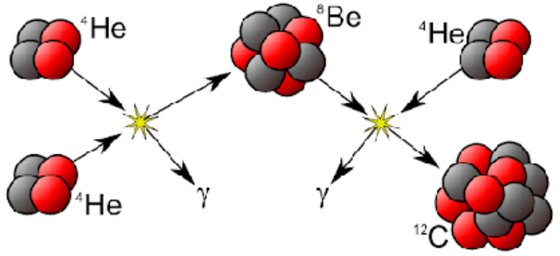
- Major uncertainty: neutrino oscillations near the core[1]
- The “oscillation problem”: simulations can’t resolve oscillations

Neutrino collective oscillations



- Quantum many-body problem
- Mean-field: 7D problem $(\vec{x}, \vec{p}, t)$
- Very short time scales[2]: ~ns
- Fast/slow, local/global simulations

Heavy-element nucleosynthesis



- Sensitive to p/n ratio (neutrinos!)
- Properties of rare isotopes unknown
- Efficient p-rich nucleosynthesis?[3,4]

[1] Long-Term Multidimensional Models of Core-Collapse Supernovae: Progress and Challenges, Annual Reviews, Hans-Thomas Janka, 2502.14836

[2] New Developments in Flavor Evolution of a Dense Neutrino Gas, Tamborra et al, Ann. Rev. Nucl. Part. Sci. 71 (2021) 165-188

[3] The p-process in supernovae, Woosley & Howard , Astrophysical Journal Supplement Series, vol. 36, Feb. 1978, p. 285-304.

[4] vp-process in core-collapse supernovae: imprints of general relativistic effects, *JCAP* 02 (2026) 067

Current challenges

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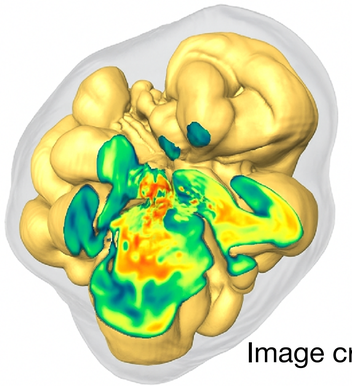
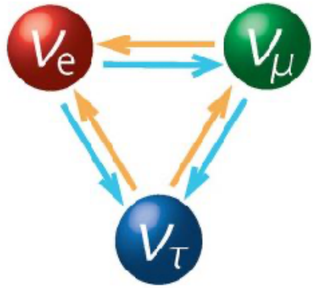


Image credits: T. Janka

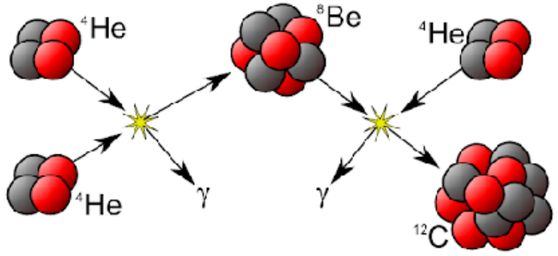
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Most urgent problem, in my opinion

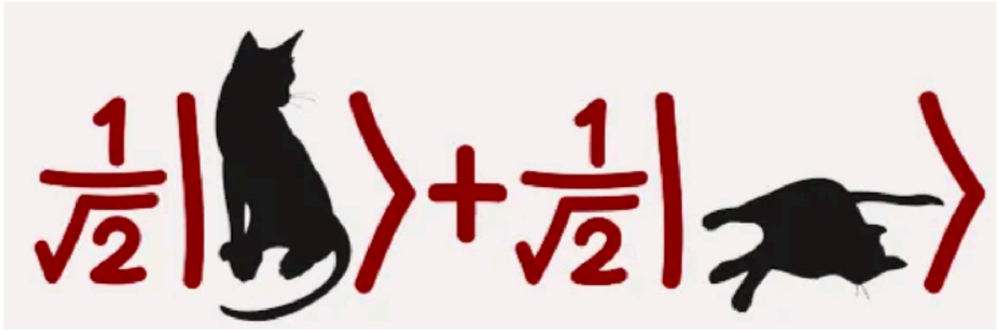
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[2] New Developments in Flavor Evolution of a Dense Neutrino Gas, Tamborra et al, Ann. Rev. Nucl. Part. Sci. 71 (2021) 165-188

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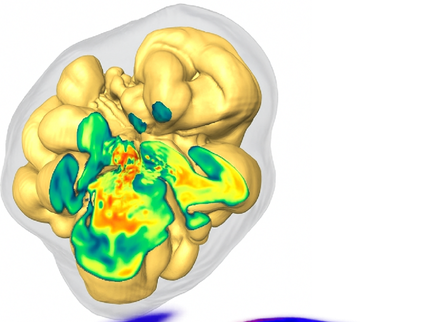
Flavor oscillations in dense media

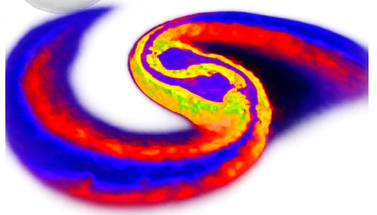


$$a_e(t)\,|\nu_e\rangle + a_\mu(t)\,|\nu_\mu\rangle$$

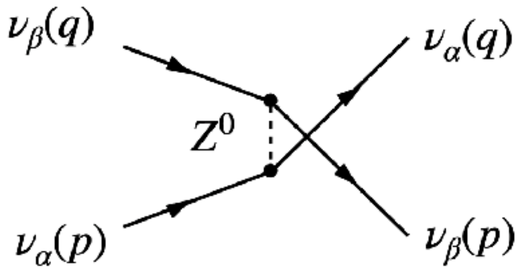
Neutrino-neutrino interaction

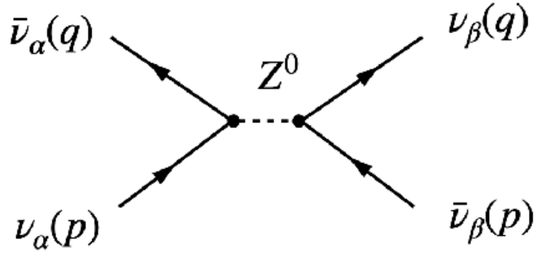
- Neutrinos also constitute a background for *other* neutrinos
- $\nu\nu$ interactions have off-diagonal elements in flavor basis[1]





Neutrino–neutrino coherent forward scattering





[1]Neutrino oscillations at high densities, Pantaleone, Physics Letters B 287 (1992) 128-132

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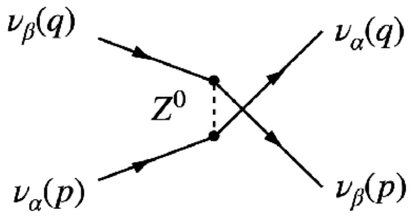
Momentum of test neutrino

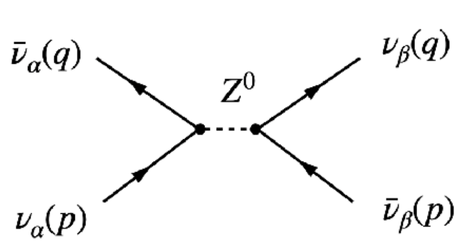
$$H_{\nu\nu}(\vec{p}) = \sqrt{2} G_F n_\nu \int d\vec{q} \left[\rho(\vec{q}) - \bar{\rho}(\vec{q})\right] \left(1 - \frac{\vec{p}\cdot\vec{q}}{|\vec{p}||\vec{q}|}\right)$$

Neutrino interaction strength

Momentum of background neutrino

Density matrices (Flavor content of the medium)





- *Non-linear* behavior
- $\nu\nu$ interaction induces *collective oscillations*[2-5]

[1]Neutrino oscillations at high densities, Pantaleone, Physics Letters B 287 (1992) 128-132

[2]Collective Neutrino Oscillations, Duan et al, Ann. Rev. Nucl. Part. Sci. 60:569 (2010)

[3]Supernova Neutrinos: Production, Oscillations and Detection, Mirizzi et al, Rivista del Nuovo Cimento Vol. 39, N. 1-2 (2016)

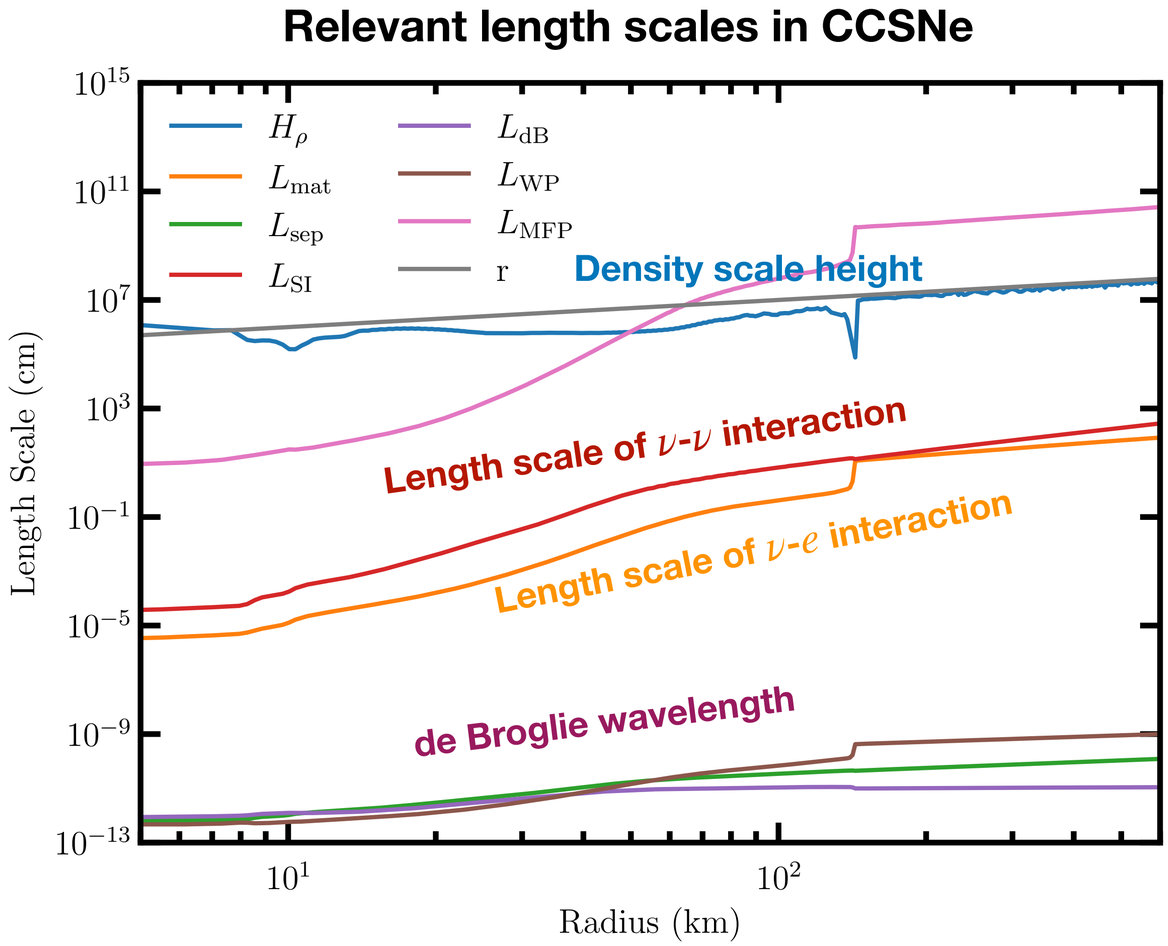
[4]Collective neutrino flavor conversion: Recent developments, Chakraborty et al, 1602.02766

[5]New Developments in Flavor Evolution of a Dense Neutrino Gas, Tamborra et al, Ann. Rev. Nucl. Part. Sci. 71 (2021) 165-188

Pedagogical resource: P. Dedin+, Mod.Phys.Lett.A 37 (2022) 08, 2250048

The oscillation problem

- Neutrinos experience a potential on **length scales much shorter** than the **macroscopic scale**
- Hydro simulations with this resolution are not possible
- The "**oscillation problem**" is to figure out how to implement flavor mixing without fully resolving it



Annual Review: L. Johns et al, 2503.05959

Quantum kinetic equations

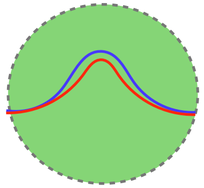
- Neutrinos with different momenta evolve collectively
- The (mean-field) equations of motion:

$$\left(\frac{\partial}{\partial t} + \vec{v}\cdot\vec{\nabla}_x\right)\rho(\vec{x},\vec{p},t) =$$

$$\left(\frac{\partial}{\partial t} + \underbrace{\vec{v}\cdot\vec{\nabla}_x}_{\text{advection}}\right)\bar{\rho}(\vec{x},\vec{p},t) =$$

Spatial variations

$$\rho = \begin{pmatrix} \rho_{ee} & \rho_{ex} \\ \rho_{ex}^* & \rho_{xx} \end{pmatrix}$$



$\alpha|\nu_e\rangle + \beta|\nu_x\rangle$

Superposition of two states

Quantum kinetic equations

- Neutrinos with different momenta evolve collectively
- The (mean-field) equations of motion:

$$\left(\frac{\partial}{\partial t} + \vec{v}\cdot\vec{\nabla}_x\right)\rho(\vec{x},\vec{p},t) = -i\left[H(\vec{x},\vec{p},t),\rho(\vec{x},\vec{p},t)\right]$$

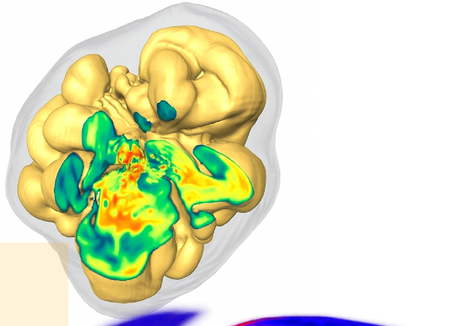
$$\left(\frac{\partial}{\partial t} + \underbrace{\vec{v}\cdot\vec{\nabla}_x}_{\text{advection}}\right)\bar{\rho}(\vec{x},\vec{p},t) = -i\underbrace{\left[\bar{H}(\vec{x},\vec{p},t),\bar{\rho}(\vec{x},\vec{p},t)\right]}_{\text{refraction}}$$

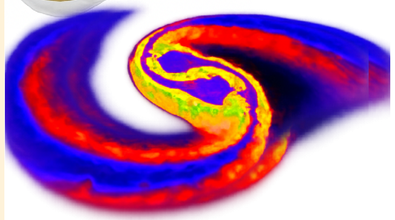
Spatial variations

Coherent forward scattering

Quantum kinetic equations

- Neutrinos with different momenta evolve collectively
- The (mean-field) equations of motion:





$$\left(\frac{\partial}{\partial t} + \vec{v}\cdot\vec{\nabla}_x\right)\rho(\vec{x},\vec{p},t) = -i\left[H(\vec{x},\vec{p},t),\rho(\vec{x},\vec{p},t)\right] + \mathscr{C}\left(\rho(\vec{x},\vec{p},t),\bar{\rho}(\vec{x},\vec{p},t)\right)$$

$$\left(\frac{\partial}{\partial t} + \underbrace{\vec{v}\cdot\vec{\nabla}_x}_{\text{advection}}\right)\bar{\rho}(\vec{x},\vec{p},t) = -i\underbrace{\left[\bar{H}(\vec{x},\vec{p},t),\bar{\rho}(\vec{x},\vec{p},t)\right]}_{\text{refraction}} + \underbrace{\bar{\mathscr{C}}\left(\rho(\vec{x},\vec{p},t),\bar{\rho}(\vec{x},\vec{p},t)\right)}_{\text{collisions}}$$

Spatial variations

Coherent forward scattering

Non-forward scattering

Very active field of research!

Recent workshop

Collective Neutrino Oscillations in Supernovae and Neutron Star Mergers

Taipei, March 23–27, 2026

- The workshop gathered researchers working on neutrino flavor conversions in explosive astrophysical environments.
- It highlighted recent progress in understanding collective neutrino effects in simulations.
- The goal was to share ideas, compare methods, and promote collaboration.



Such a few examples:

- **Marie Cornelius (Dortmund U)** — Diagnosing ELN Crossings in Supernovae: *ELN crossings in core-collapse supernovae are strongly shaped by proto-neutron-star convection and weakly by the equation of state or muons, with convection pushing crossings to larger radii and muons to smaller radii.*

- **Ryuichiro Akaho (Waseda U)** — Fast Flavor Conversion in CCSNe: *Fast neutrino flavor conversion can significantly alter supernova heating and explosion dynamics, but simplified moment-based transport methods tend to underestimate ELN crossings and flavor-instability effects.*

- **Tianshu Wang (UC Berkeley)** — Collective flavor oscillations in CCSN simulations: *CFI is negligible and FFC alters neutrino signals but barely affects CCSN dynamics.*

- **Kanji Mori (Keio University)** — 3D CCSN models with phenomenological flavor conversions: *Phenomenological flavor conversions help SN explosion, enhancing explosion energy and shock revival in 3D.*

- **Hiroki Nagakura (Kyoto U)** — Neutrino Quantum Kinetics in SNe & BNSMs: *Subgrid models (e.g., BGK) are essential for including flavor conversions in CCSN/BNSM simulations; global QKE results can differ qualitatively from local ones.*

- **Soumya Bhattacharyya (Academia Sinica)** — FFC: Role of Matter Inhomogeneity: *Strong matter-potential gradients suppress FFC; a simple m vs. m_crit criterion can decide whether FFC should be included in SN simulations.*

- **Masamichi Zaizen (Univ. of Tokyo)** — FFC with attenuation and global lepton gradient: *Apparent suppression of FFI by global radial gradients in background leptons can be artificial — strong attenuation and insufficient resolution distort the result in global QKE simulations, especially for CFI/SFI.*

- **Damiano Fiorillo (INFN Napoli)** — Theory of neutrino flavor plasma: *Treating flavor waves as real plasma excitations unifies fast/slow/collisional instabilities through Feynman-rule growth rates, with saturation set by detailed balance and "flavomon" emission heating/cooling the neutrino plasma.*

- **Jiabao Liu (Waseda U)** — Linear analysis and nonlinear asymptotics of flavor conversions: *Proposed a pole-density "Method C" that fixes divergences in CFI dispersion relations, and showed QuasiLinear phase synchronization reproduces QKE asymptotic states.*

- **Anson Kost (UNM)** — Quasi-steady states and miscidynamics in OILE models: *Entanglement rate scales with a wavepacket-size parameter γ, and miscidynamics (adiabatic evolution between mean-field equilibria) reproduces the many-body dynamics.*

- **Yi Qiu (Penn State)** — Subgrid neutrino flavor mixing in BNS mergers: *Flavor conversions boost heavy r-process yields by ~200–1000%, yield more compact remnants, and shift post-merger GW peaks by ~100 Hz.*

- **David Radice (Penn State)** — Neutron-rich outflows from NS mergers and AICs: *Secular winds dominate BNS ejecta and r-process yields appear nearly solved, but cracks remain from long-term dynamics, MHD, neutrino uncertainties, and LGRB-kilonovae hinting at additional sources.*

- **Heamin Ko (TU Darmstadt)** — Heavy-flavor neutrinos in the νr-process: *NC reactions from νμ/ντ boost p-nuclei production for ⟨E_νx⟩ ≲ 41 MeV, but above ~44 MeV α-spallation flips the outcome to an r-process-like pattern, with ^{17}O as the key bottleneck.*

- **Harry Ho-Yin Ng (Goethe U)** — Muonic neutrino transport in BNS mergers: *Adding muonic β-processes via 5-species transport (Weakhub) suppresses electronization, yielding less proton-rich ejecta and introducing new muonic flavor-oscillation sectors relevant for kilonova/r-process modeling.*

- **Noshad Khosravi Largani (University of Wroclaw)** — QCD Phase Transitions in CCSNe: *QCD phase transitions in failed supernovae create a second shock and distinctive neutrino burst linked to explosion and black-hole formation.*

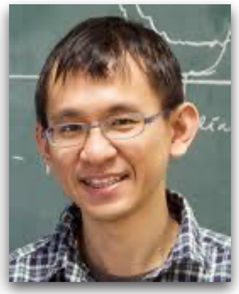
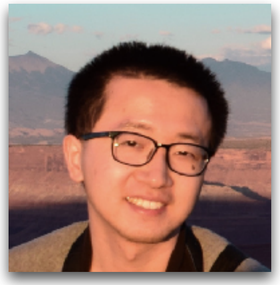
- **Madhurima Chakraborty (Academia Sinica)** — ML Detection of ELN Crossings: *Machine-learning methods can robustly identify non-axisymmetric ELN crossings from limited neutrino transport data.*

Flavor Equilibration of Supernova Neutrinos: Exploring the Dynamics of Slow Modes

arXiv: 2505.11588 (Phys. Rev. D 112, 043039)

In collaboration with:
Heng-Hao Chen (Jordan), Sajad Abbar, Meng-Ru Wu, and Zewei Xiong



Hamiltonian of the system

- **Vacuum Hamiltonian**

$$H_{\rm vac} = \frac{\omega}{2}\begin{pmatrix} -\cos 2\theta_V & \sin 2\theta_V \\ \sin 2\theta_V & \cos 2\theta_V \end{pmatrix}$$

$\omega = \dfrac{\Delta m^2}{2E}$, matter-suppressed $\theta_V = 10^{-5}$ rad

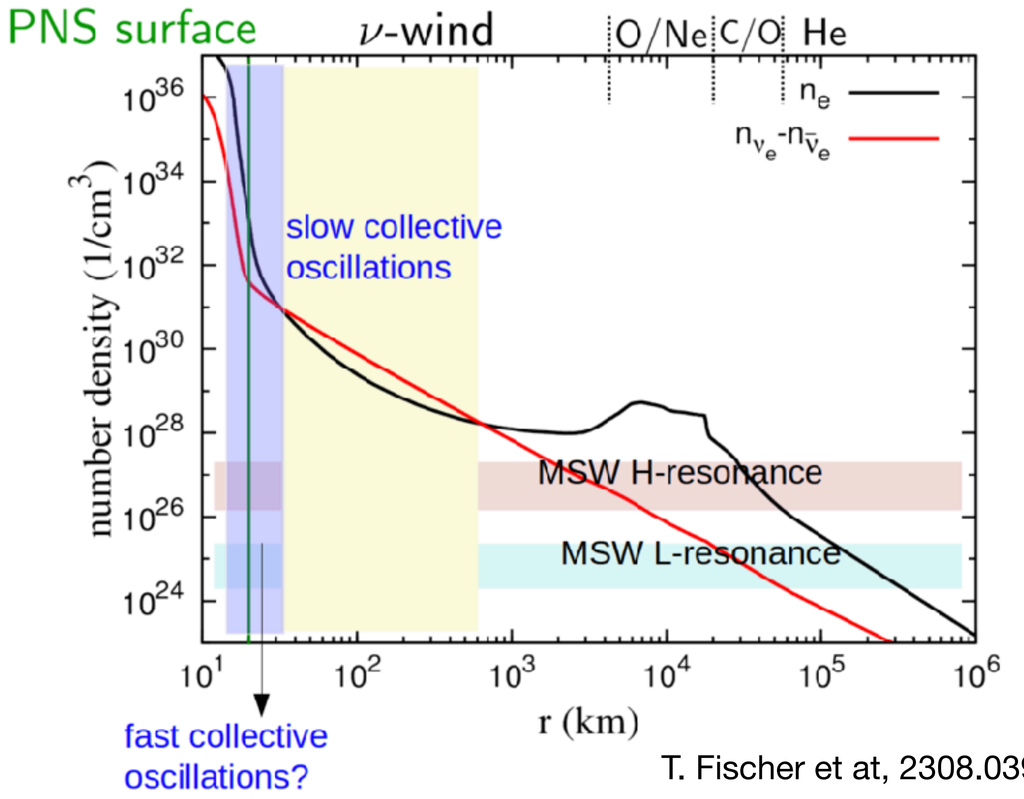
(single-energy $E = 10$ MeV$\rightarrow \omega \simeq 0.6$ km^{-1}

- **Neutrino-neutrino forward scattering**

$$H_{\nu\nu} = \mu \int dv_z' [\rho(t, z, v_z') - \bar{\rho}(t, z, v_z')](1 - v_z v_z')$$

$\mu \equiv \sqrt{2} G_F n_{\nu_e}$: self-interaction potential

(assume $n_{\nu_e} \simeq 10^{29}$ cm$^{-3} \rightarrow \mu \simeq 64.2$ km^{-1}, ratio $\omega/\mu \approx 0.01$)



T. Fischer et at, 2308.03962

Neutrino gas in a box

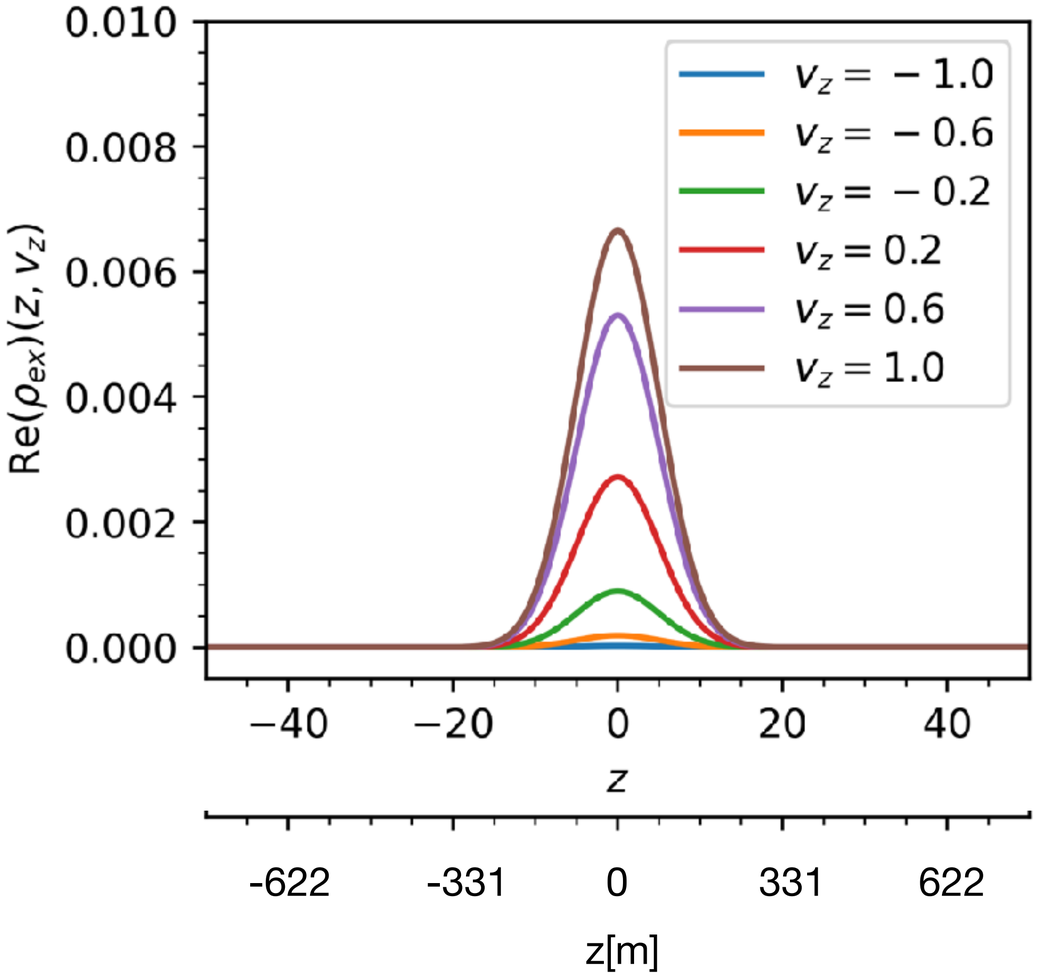
Initial configuration in space

- Box size: $L = 100[\mu^{-1}] \simeq 1.5\ \mathrm{km}$
- Periodic boundary (local quasi-equilibrium)
- Initial z-dependence of seed

$$\rho_{ex}(t=0) \propto e^{-z^2/50} \text{ (i.e. width } \simeq 77\ \mathrm{m})$$

- Initial size of inhomogeneity

$$\frac{\rho_{ex}(z, v_z, t=0)}{\rho_{ee}(z, v_z, t=0)} < 10^{-2}$$



Neutrino gas in a box

Initial velocity profile

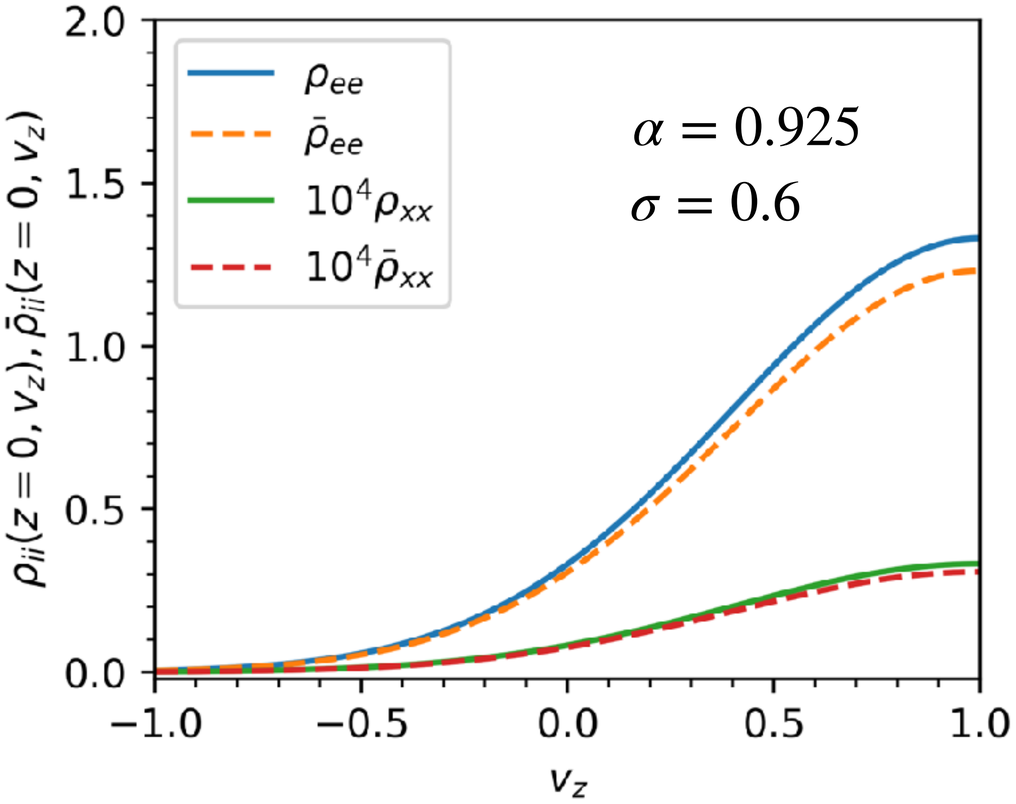
- Velocity bins within $-1 < v_z/c < 1$
- Forward-peaked angular profile (no ELN-XLN crossings)

$$g(v_z, \sigma) \propto e^{-(v_z-1)^2/(2\sigma^2)}$$

- Antineutrino-neutrino number density ratio

$$\alpha \equiv n_{\bar{\nu}_e}/n_{\nu_e}$$

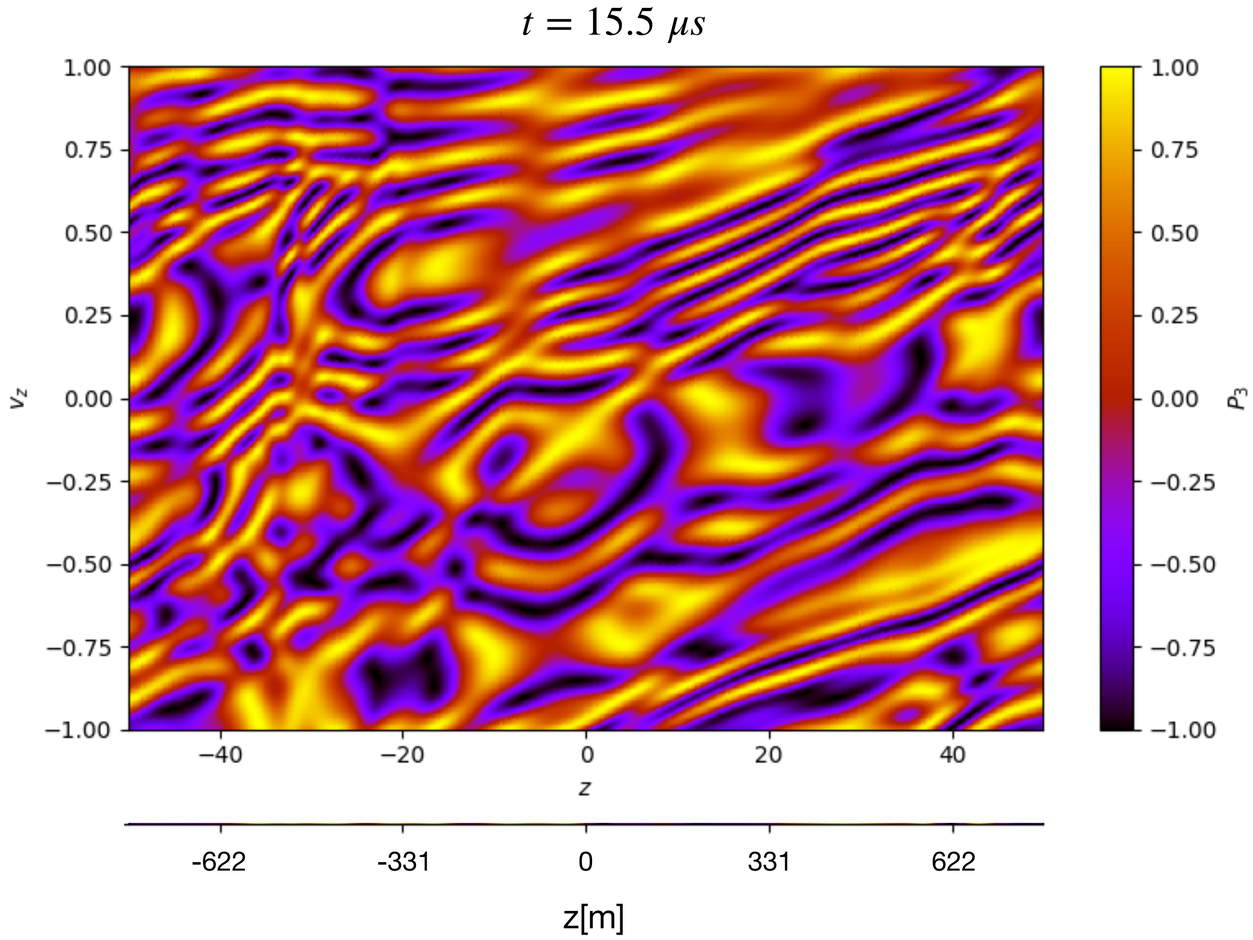
- Effectively no ν_x, $\bar{\nu}_x$ flavors



Neutrino gas in a box

Method

- Solve nonlinear PDEs with COSEν [M. George et al, 2203.12866]
- Discretization z (1000 bins)
- Discretization v_z (512 bins)
- Advection: Finite difference method
- Time evolution: RK4



I. Padilla-Gay et al, 2505.11588

https://github.com/COSEnu/COSEnu

Early dynamics

Linear stability analysis

- Early dynamics ($t \ll 1/\omega$) described by ρ_{ex}

$$\rho = \begin{pmatrix} \rho_{ee}(t_0) - \Delta(t) & \rho_{ex}(t) \\ \rho^*_{ex}(t) & \rho_{xx}(t_0) + \Delta(t) \end{pmatrix}$$

Diagonal terms evolve as $\Delta \sim \mathcal{O}(\rho_{ex}^2)$

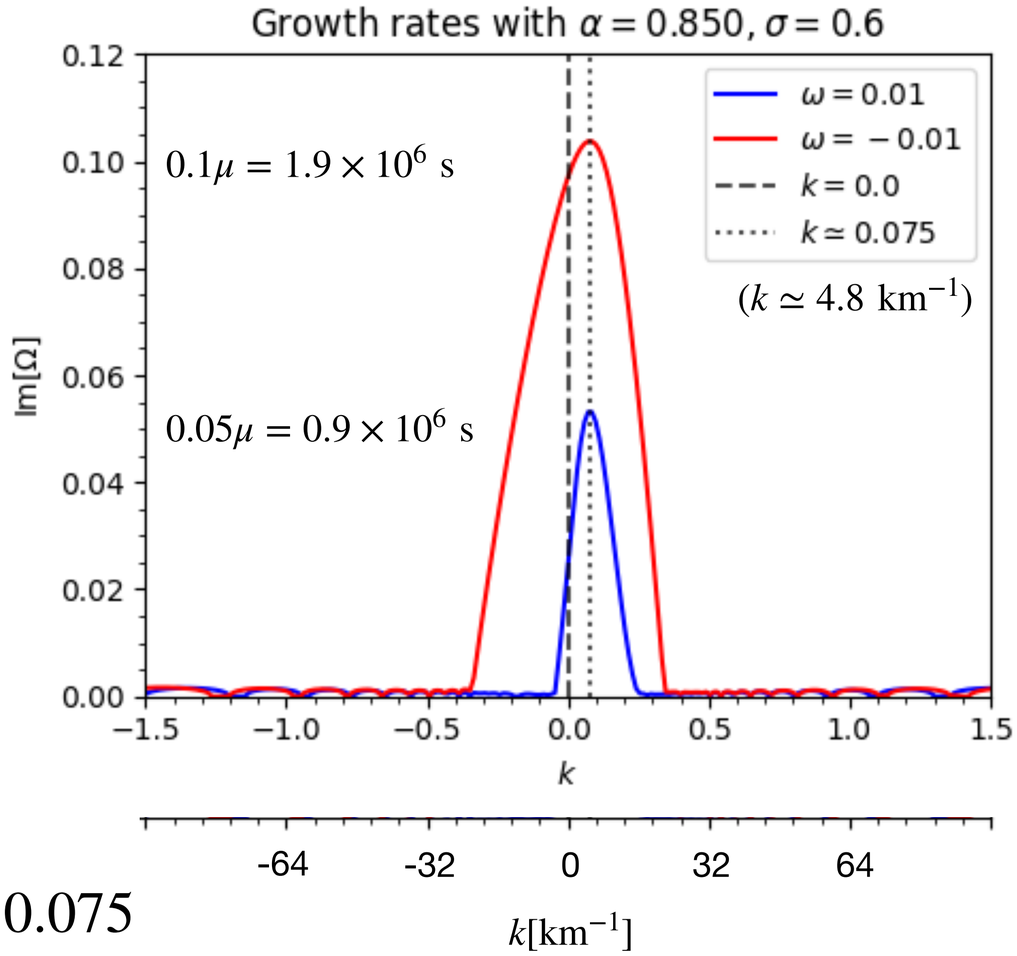
- Dynamics of off-diagonal terms

$$\rho_{ex}(v_z) \propto \sum_i C_j e^{-i(\Omega_j t - k_j z)}$$

$\mathrm{Im}[\Omega](k)$: timescale, box size and resolution required

"Zero mode": $K = k - \int dv v(\rho_{ee} - \bar{\rho}_{ee}) = 0 \rightarrow k \simeq 0.075$

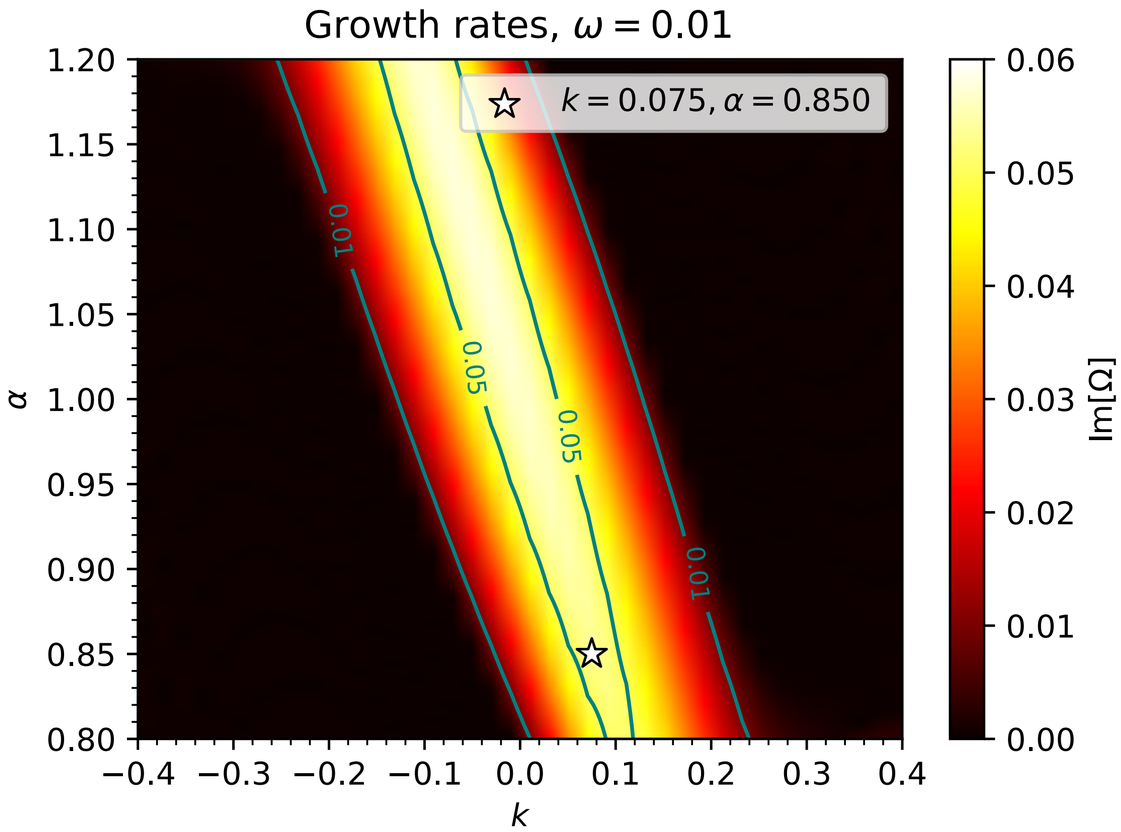
(If isotropic: $k \simeq 0$)

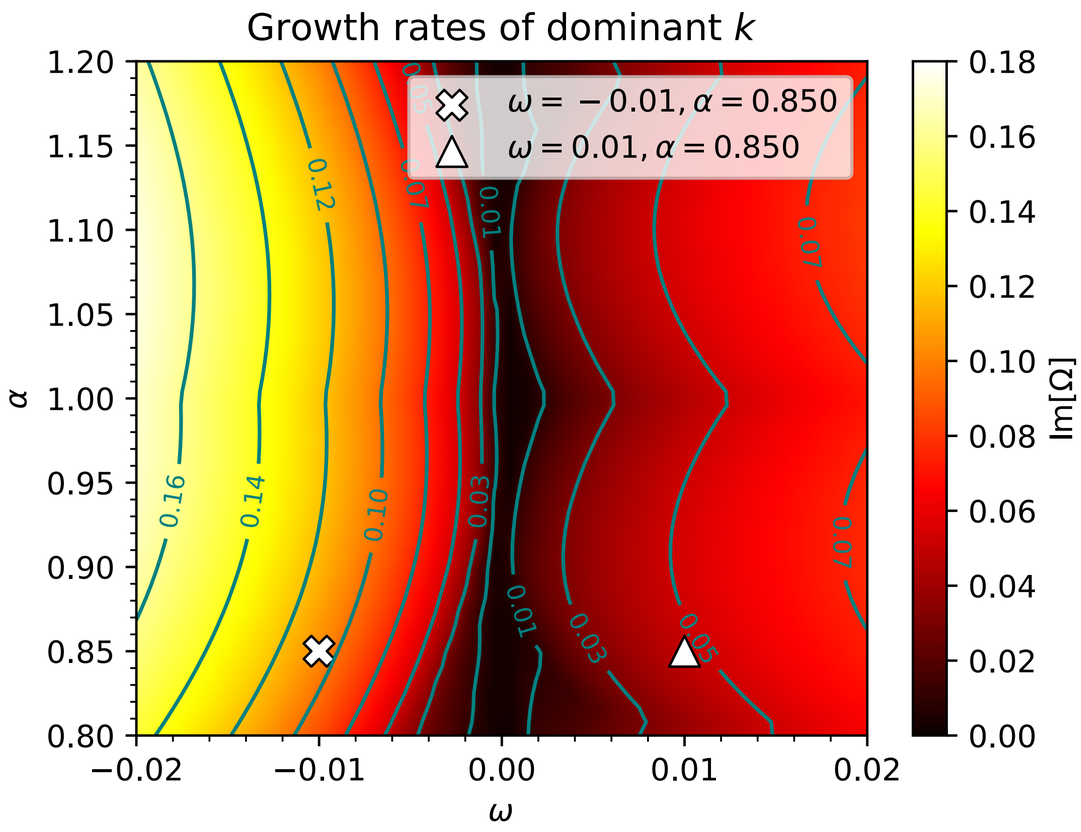


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Early dynamics

Linear stability analysis

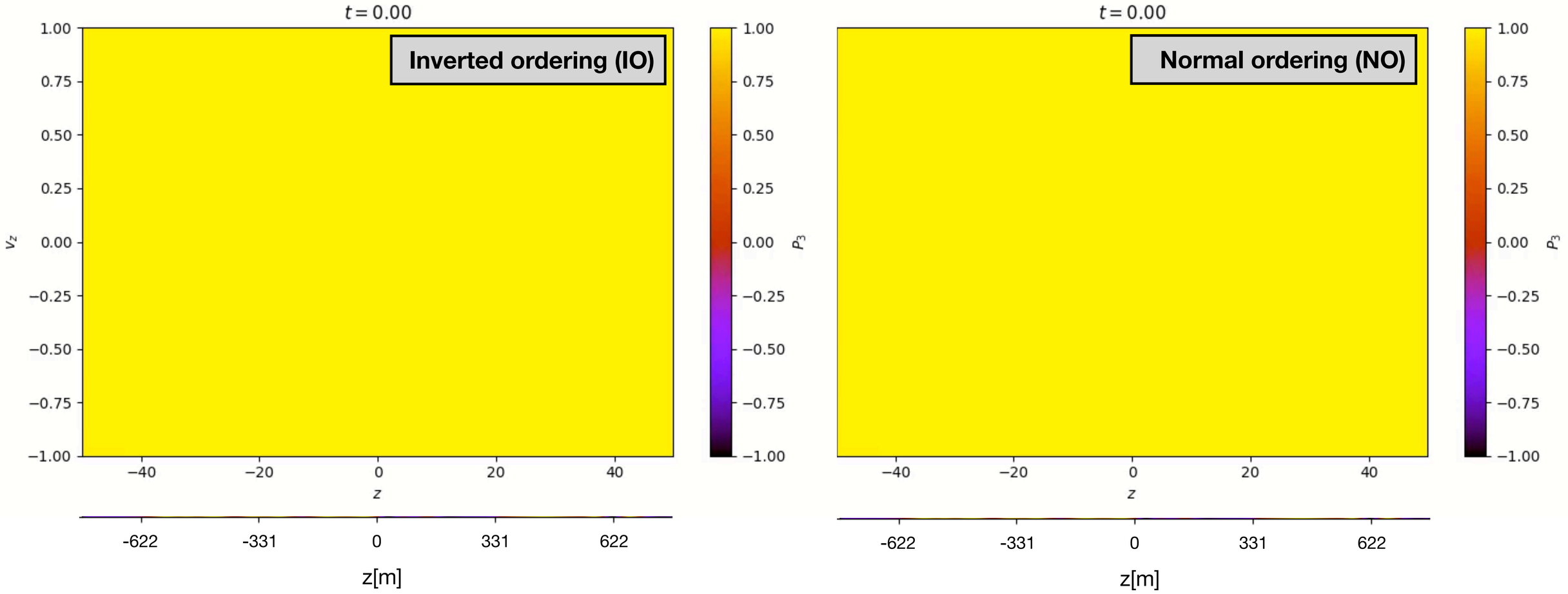




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Flavor evolution: coarse-grained study

Final $t \simeq 52\ \mu s$



sorry this is supposed to be a movie not visible in pdf (ask me for them if you're curious)

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Flavor evolution: coarse-grained study

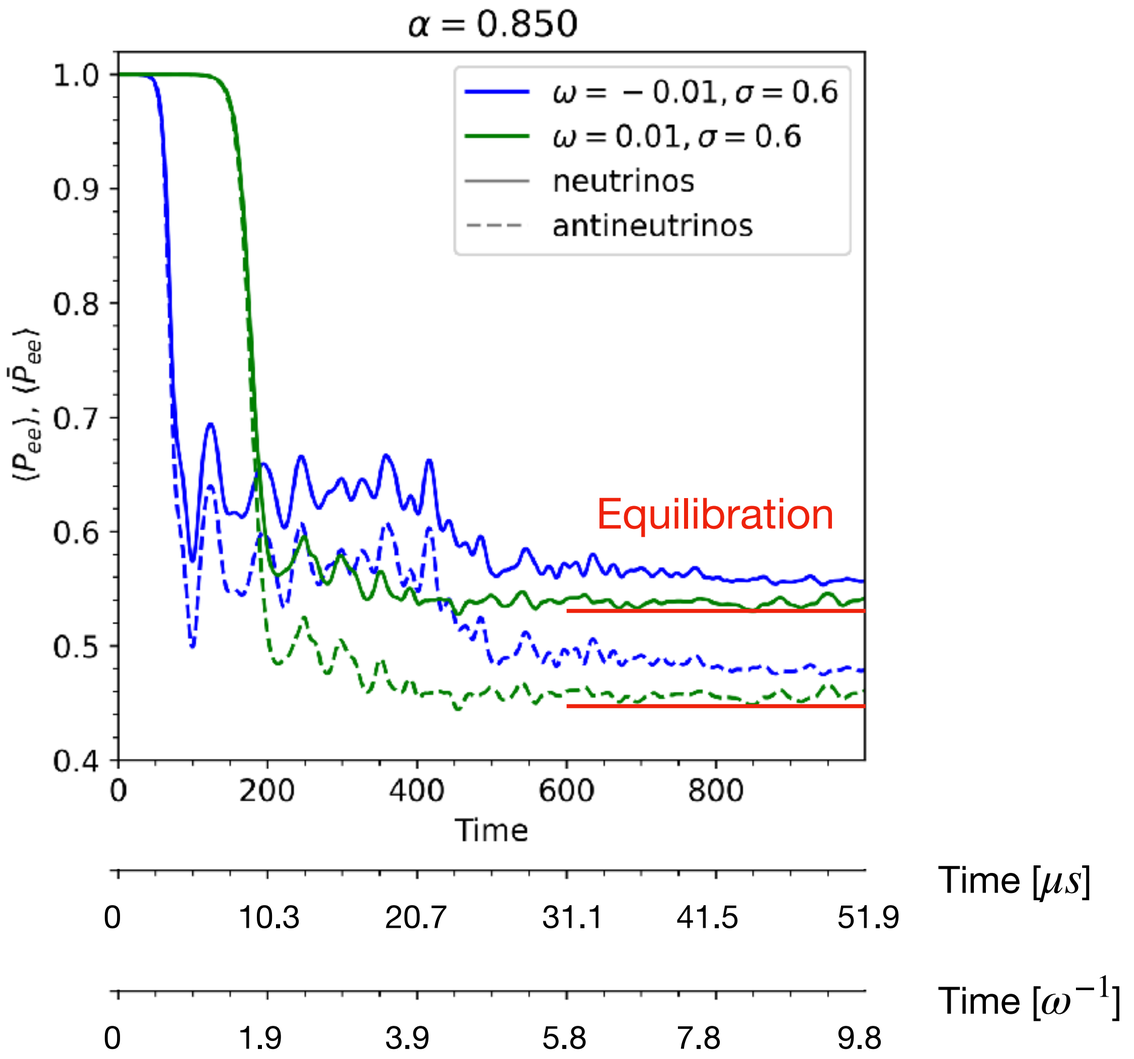
- Coarse-grained probability

$$\langle P \rangle_{z,v_z}(t) = \frac{\int dz dv_z \rho_{ee}(z, v_z, t)}{\int dz dv_z \rho_{ee}(z, v_z, t=0)}$$

- Flavor equilibration reached within the timescale $\sim 6\omega^{-1}$ (i.e. 31 μs)
- Conservation of lepton number requires relation between probabilities

$$1 - \langle P \rangle_{z,v_z} \sim \alpha(1 - \langle \bar{P} \rangle_{z,v_z})$$

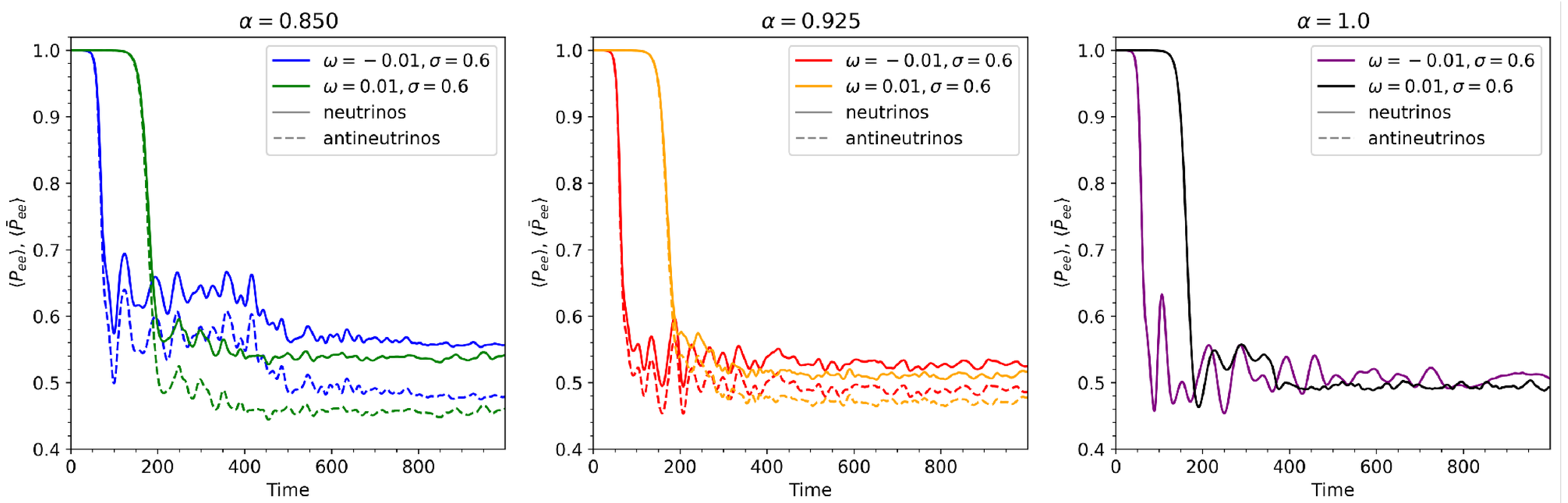
(remember $\alpha = n_{\bar{\nu}_e}/n_{\nu_e}$)



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Flavor evolution: coarse-grained study

- Coarse-grained probability



$$1 - \langle P \rangle_{z,v_z} \sim \alpha(1 - \langle \bar{P} \rangle_{z,v_z})$$

(remember $\alpha = n_{\bar{\nu}_e}/n_{\nu_e}$)

0 10.3 20.7 31.1 41.5 51.9 Time $[\mu s]$

0 1.9 3.9 5.8 7.8 9.8 Time $[\omega^{-1}]$

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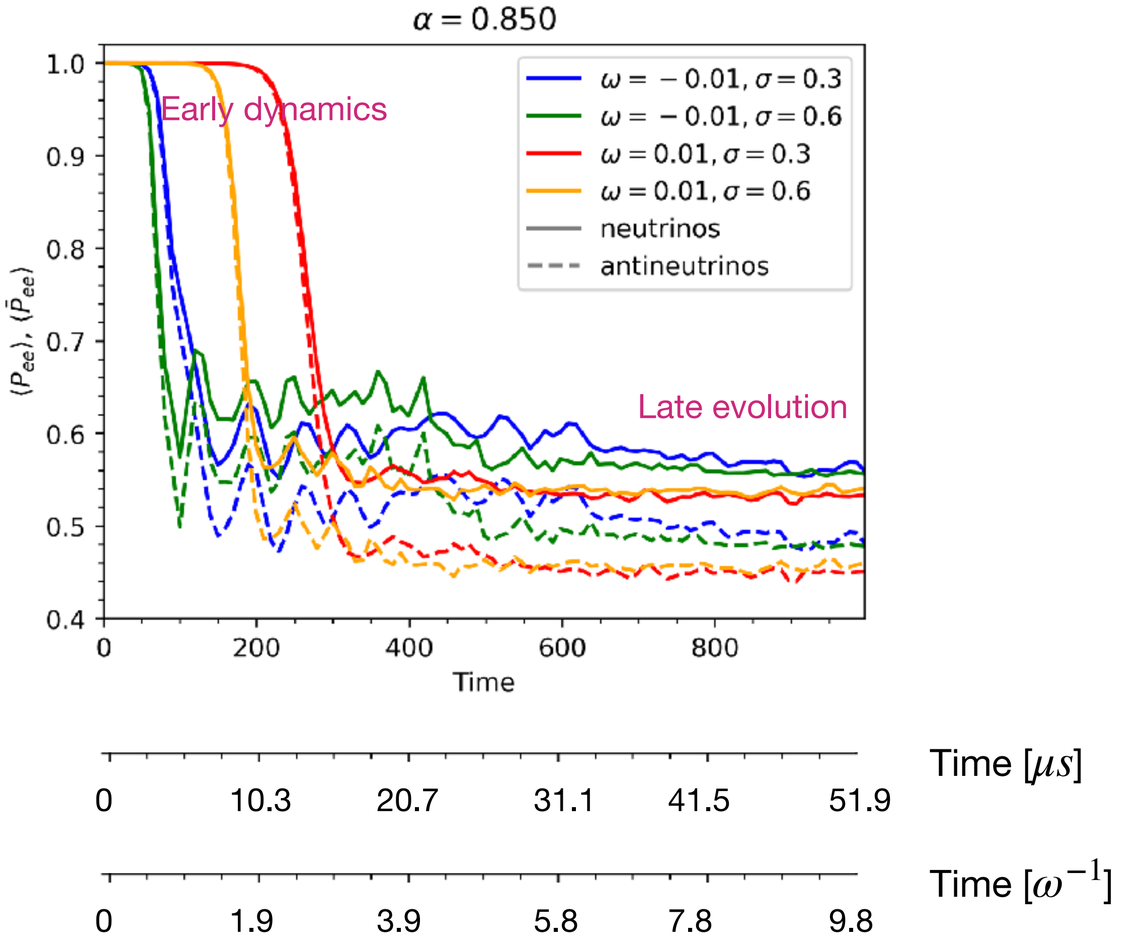
Flavor evolution: coarse-grained study

Early dynamics

- Depends on angular distributions
- Faster growth rate in IO than NO

Late evolution

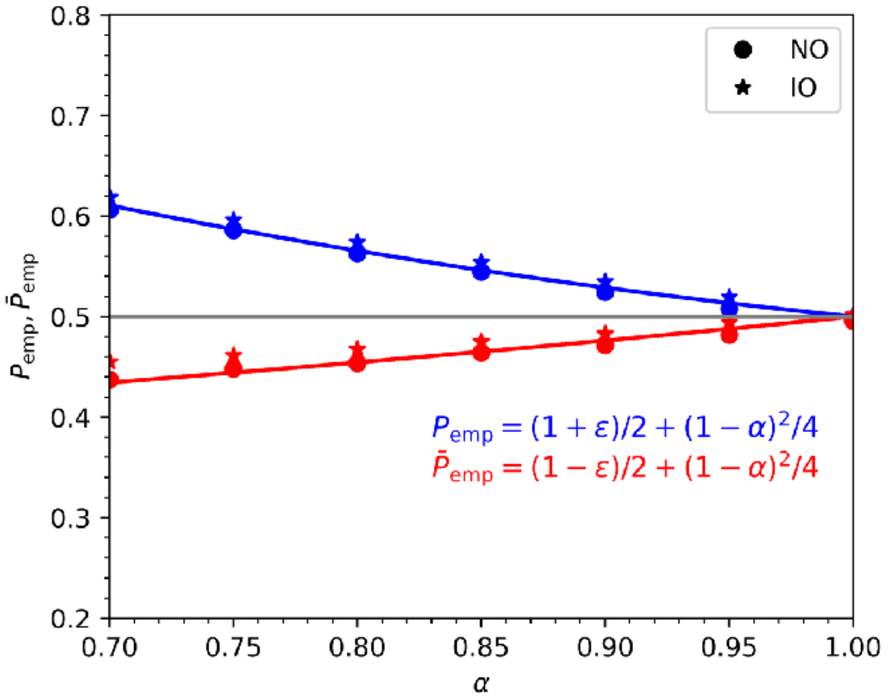
- Doesn't depend on angular distributions
- Marginally depends on mass ordering (next slides)

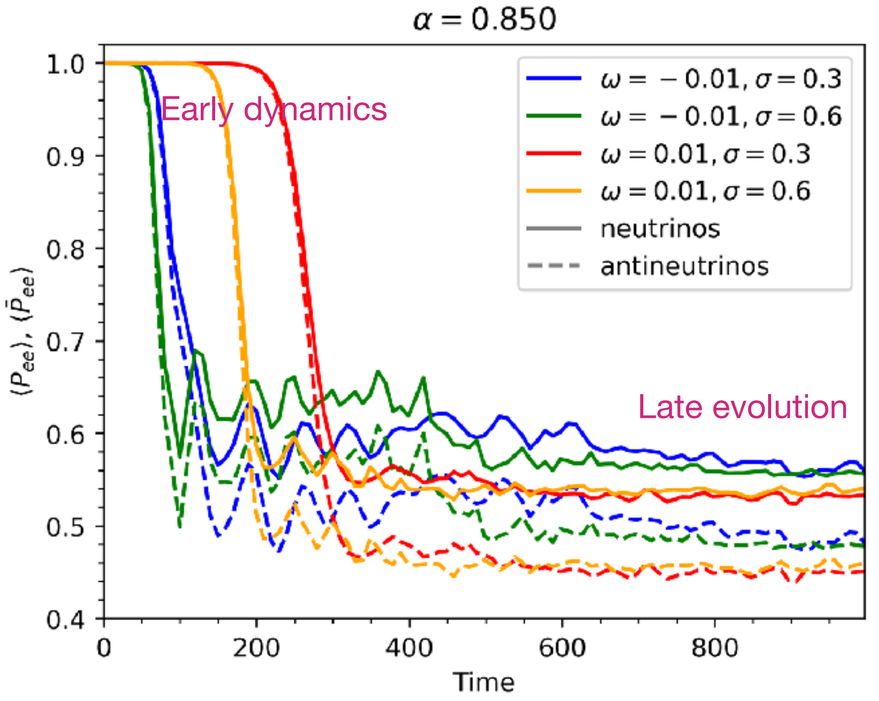


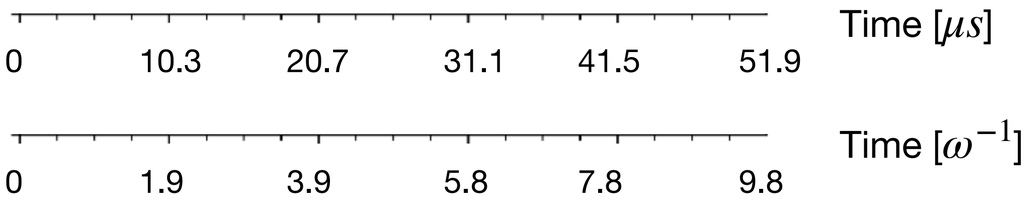
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Flavor evolution: coarse-grained study

Empirical estimation of final state







ratio between lepton and particle number (see Fiorillo et al 2412.02747)

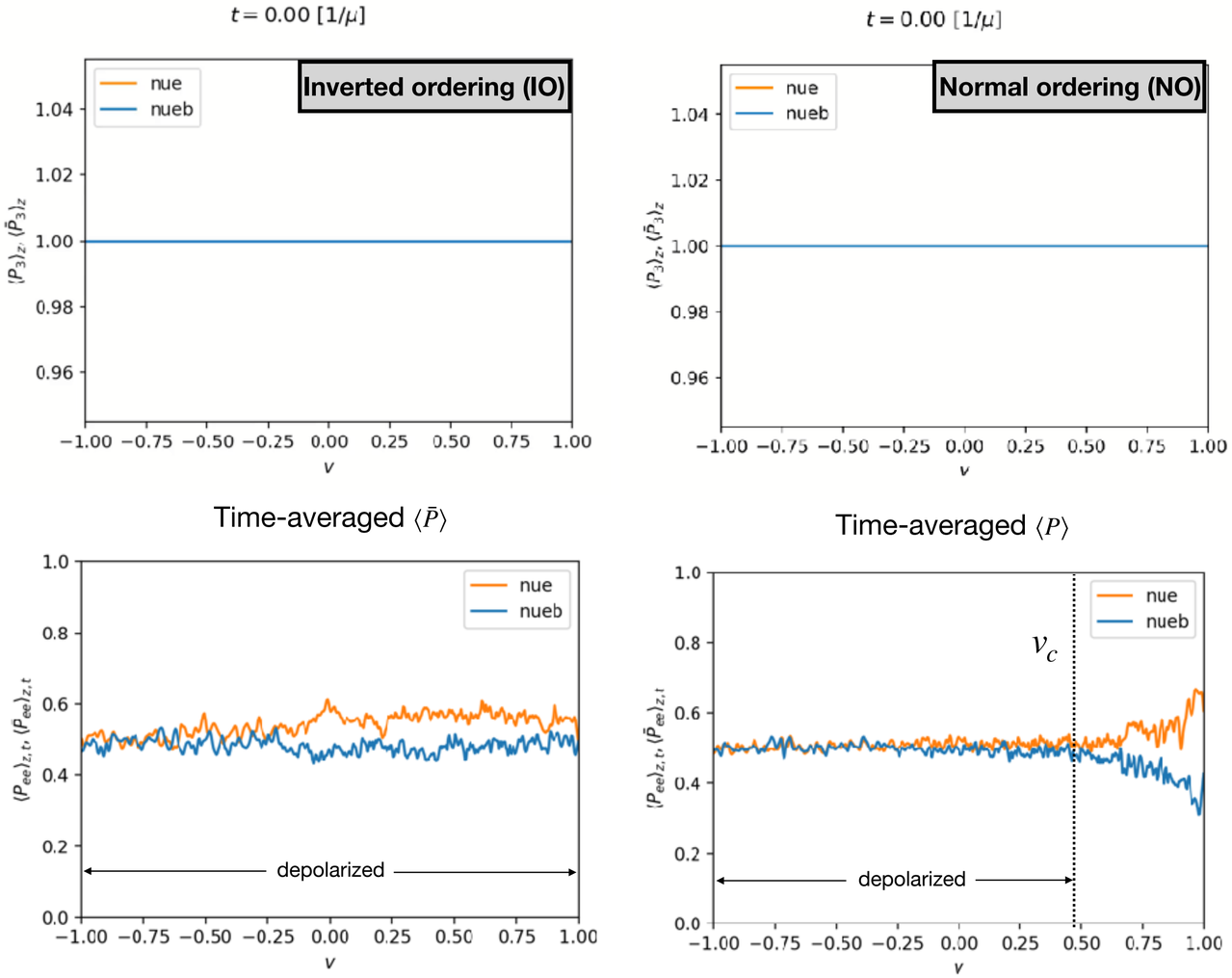
$$\epsilon \equiv |1-\alpha|/(1+\alpha)$$

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Flavor evolution: coarse-grained study

The time-averaged angular distributions depends on the mass ordering:

- In IO, deviation from flavor equilibration depends less on the velocity.
- In NO, for $v < v_c$: **completely depolarized**.
- In NO, for $v > v_c$: the deviation from flavor equilibration becomes more **pronounced***.



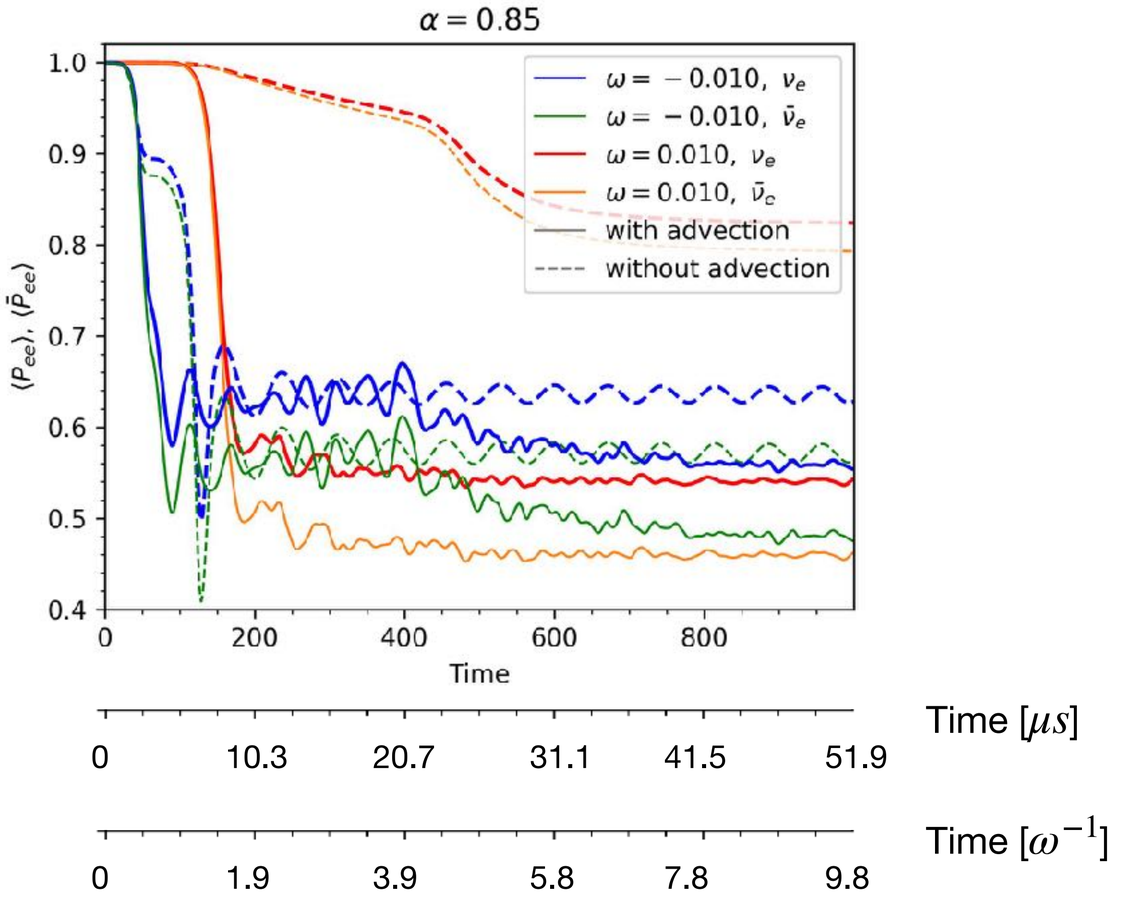
* v_c can be explained by looking at the eigenfunctions from the LSA

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Flavor evolution: coarse-grained study

Role of neutrino advection

- It strongly influences the quasi-steady sate
- Normal ordering more severely affected (remember $\omega > 0$ case dominated by non homogeneous modes)



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Main takeaways

- Dense neutrino gases in astrophysical environments exhibit nonlinear collective flavor evolution that cannot be resolved in global simulations
- Quantum kinetic calculations show that unstable systems very often evolve toward flavor equipartition
- Quasi-steady states can be estimated based on the ratio of number densities
- Multi-energy calculations within the same framework are under investigation